

CY 558 : Inorganic Chemistry for Sustainability

Course Code : CY 558

Course Name : Inorganic Chemistry for Sustainability

L-P-T-C: 3-0-0-3

Intended for: PG and PhD Level

Prerequisites: None

Mutual Exclusion: None

Approval : 57th BoA

Course Contents

- Introduction: Introduction to sustainability, life cycle sustainability assessments, social dimensions of sustainability, life cycle costing in sustainability assessment-a case study of remanufactured alternators, valorization, greenhouse gases and greenhouse effect, greenhouse gas emission associated with primary metal production, recycling, downcycling and upcycling of metals, modern trends in inorganic analysis. (5 hours)
- Solvent systems for sustainability: Bio-derived solvents, water as a solvent, liquid N₂, supercritical CO₂, CO₂ expanded liquids, CO₂ switchable solvents. (5 hours)
- Inorganic waste treatment: Recovery of gold from sewage sludge, recovery of gold carbon-based materials, carbon in pulp method: a case study from Japan, earth recycling from neodymium-iron-boron permanent magnets, development of recycling technology for rare earth metals by Hitachi: case study, adsorption methods for metal removal. (6 hours)
- Sustainable water remediation: Reasons and problems related to wastewater, bioremediation, zeolites as treatment agents for wastewater, modified silica gels as chelating sorbents, ionic liquids for water remediation, inorganic hybrid materials for water remediation, adsorption methods for water treatment, water treatment by electrocoagulation. (9 hours)
- Sustainable inorganic catalysts for organic transformations: Inorganic catalyst (Re, Ti, Mn, W) for epoxidation, catalytic epoxidation of oils, fatty derivatives and terpenes, (Al, K and Fe) based catalysts for carbonate synthesis, fluorine catalysts for hydrocarbon oxidation. Metal oxide (CuO, RuO₂, WO₃, CeO₂) supported catalytic transformation involving amination, electrophilic cyclization, C-C coupling and hydrogenation. (7 hours)
- Sustainable synthesis with microwave irradiation: Microwave assisted controlled organic synthesis, microwave assisted polymerization, synthesis of inorganic solids using microwaves, microwave assisted nanoparticle synthesis. (5 hours)

Text books:

1. David A. Atwood (Editor), **Sustainable Inorganic Chemistry**, Wiley, 2016.
2. F. Cavani, G. Centi, S. Perathoner, and F. Trifiro, **Sustainable Industrial Chemistry**, Wiley-VCH Verlag, 2009.



References:

1. Introduction: Sustainable Chemistry, **Chemical Reviews**, 118, 369–371, 2018.
2. J. J. M Nelson; E. J. Schelter, Sustainable Inorganic Chemistry: Metal Separations for Recycling, **Inorganic Chemistry**, 58, 979–990, 2019.
3. The Materials Science behind Sustainable Metals and Alloys, **Chem. Rev.**, 123, 2436–2608, 2023.
4. New Technology for Gold Recovery, JOGMEC report, Japan Oil, Gas and Metal National Corporation (JOG-MEC), Tokyo, 1995.
5. Insight; <http://www.51report.com/research/3051099.html>
6. E. M. Schau, M. Traverso, A. Lehmann and M. Finkbeiner, **Sustainability**, 3, 2268, 2011.